

56-399-650<sup>Print</sup>

## Engine Test Certificate

Perkins Engine Company Limited  
Eastfield, Peterborough PE1 5FQ  
Great Britain

May 22, 2018



This engine has been tested to its maximum rating in accordance with Perkins defined requirements and has performed satisfactorily.

The electronics on this engine have been preset and when the engine has been run-in, the gross power output will be within +/-3% of the gross rating listed below, provided the exhaust and induction systems, fuel and lubricating oils are in accordance with Perkins defined requirements.

| Engine Type   | Perkins Serial Number | Customer |
|---------------|-----------------------|----------|
| 2806A-E18TAG2 | JGBF5005N04813D       | LPS2000  |

| Reference                        | Technical Data      |
|----------------------------------|---------------------|
| Emission Capability              | Unregulated Low SFC |
| Engine Speed                     | 50 Hz @1500RPM      |
| Engine Type                      | 2806A-E18TAG2       |
| Fuel Consumption - Prime Power   | 202 g/kWh           |
| Fuel Consumption - Standby Max   | 203 g/kWh           |
| Gross Engine Power - Prime Power | 584 kW              |
| Gross Engine Power - Standby Max | 628 kW              |
| Gross Rating                     | 584 kW              |
| Net Rating                       | 565 kW              |

This engine was built and tested in accordance with our approved ISO 9001 Quality Procedure and stated Rating, as determined by the ISO 3046 (BS5541) Standard.

Performance certified as satisfactory.

Signed,

For and on behalf of Perkins Engines Company Limited - Large Engines Division.

**MUST BE KEPT WITH ENGINE AT ALL TIMES**

Issue 4

# Test Certificate

FOR  
SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATOR  
Machine No. **N19G304489**

Type Test has been conducted on similar design machine  
STANDARDS: Generator generally conforms to BS 5000 : Part 3,  
Tested in accordance with BS EN 60034-1/IS/IEC 60034-1  
Rotor dynamically balanced to BS ISO 1940 - 1

## ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:

|                  |                       |                      |                       |              |
|------------------|-----------------------|----------------------|-----------------------|--------------|
| FRAME: HCI544F1  | AVR: AS440            | AVR NO: 191401480069 | ENCLOSURE: IP23       |              |
| RATED KVA: 650   | RATED VOLTS: 415      | RATED AMPS: 904.3    | FREQUENCY: 50         | PHASE: 3     |
| POWER FACTOR: .8 | STR CONN: Series Star | STR WINDING: 311     | DUTY: Base Continuous | WIRE: 4-Wire |
| EXC. VOLTS: 48   | EXC.AMPS: 2.3         | AMB TEMP: 40         |                       |              |

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

| WINDING | COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES) | INSULATION RESISTANCE (M- Ohm) | HV TEST 2KV FOR 1 MIN | INSULATION CLASS |
|---------|---------------------------------------------------------------|--------------------------------|-----------------------|------------------|
| STATOR  | 0.0078                                                        | 2+                             | OK                    | H                |
| ROTOR   | 2.284                                                         | 2+                             | OK                    | H                |

## REGULATION TEST

| FREQ. | VOLTS | AMPS  | KVA  | PF   | KW   | EXC.VOLT | EXC.AMP | % LOAD |
|-------|-------|-------|------|------|------|----------|---------|--------|
| 52    | 416.4 | 0.00  | 0.00 | 0.00 | 0.00 | 10       | .54     | 0.00   |
| 50    | 415   | 904.3 | 650  | .8   | 520  | 48       | 2.3     | 100.00 |

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)\*100 = 0.34 %

Item Description: AC Generator.HC5.4.F.1.311.,0,,18,AS440.Black - RAL9011  
Remarks:

|                 |                     |                                                                |
|-----------------|---------------------|----------------------------------------------------------------|
| Date: 25-JUL-19 | Job No: 3749706-001 | AUTHORISED BY<br><b>Manoj K Upadhyaya</b><br>Quality Assurance |
|-----------------|---------------------|----------------------------------------------------------------|

## Certificate Of Conformity

Quality System Certification  
ISO 9001:2015

Frame Size : **HCI544F1**

Serial Number : **N19G304489**

KVA: **650**

Manufactured Date : 25-JUL-19

Authorized By Quality Assurance: **Manoj K Upadhyaya**

This document is electronically generated, hence signature not required.

Print Date: 25-JUL-19

# EU DECLARATION OF CONFORMITY



This synchronous A.C. generator is designed for incorporation into an electricity generating-set and fulfils all the relevant provisions of the following EU Directive(s) when installed in accordance with the installation instructions contained in the product documentation:

|            |                                                                                             |
|------------|---------------------------------------------------------------------------------------------|
| 2014/35/EU | Low Voltage Directive                                                                       |
| 2014/30/EU | The Electromagnetic Compatibility (EMC) Directive                                           |
| 2011/65/EU | Restriction on Hazardous Substances in Electrical and Electronic Equipment (RoHS) Directive |
| 2015/863   | Delegated Directive amending Annex II of 2011/65/EU                                         |

and that the standards and/or technical specifications referenced below have been applied:

|                           |                                                                                                                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EN 61000-6-2:2005         | Electromagnetic compatibility (EMC). Generic standards - Part 6-2: Immunity for industrial environments                                                                                                   |
| EN 61000-6-4:2007+A1:2011 | Electromagnetic compatibility (EMC). Generic standards - Part 6-4: Emission standard for industrial environments                                                                                          |
| EN ISO 12100:2010         | Safety of machinery - General principles for design - Risk assessment and risk reduction                                                                                                                  |
| EN 60034-1:2010           | Rotating electrical machines - Part 1: Rating and performance                                                                                                                                             |
| BS ISO 8528-3:2005        | Reciprocating internal combustion engine driven alternating current generating sets - Part 3: Alternating current generators for generating sets                                                          |
| BS 5000-3:2006            | Rotating electrical machines of particular types or for particular applications - Part 3: Generators to be driven by reciprocating internal combustion engines - Requirements for resistance to vibration |
| EN 50581:2012             | Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances                                                                  |

This declaration has been issued under the sole responsibility of the manufacturer. The object of this Declaration is in conformity with the relevant Union harmonization Legislation.

The name and address of authorised representative, authorised to compile the relevant technical documentation, is the Company Secretary, Cummins Generator Technologies Romania, B-dul Decebal Nr. 116A 200746 Craiova Dolj, Romania.

Signed:

Date:

6<sup>th</sup> March 2019

Name, Title and Address:

Kevan J Simon  
Global Technical Director  
Cummins Generator Technologies Romania  
B-dul Decebal Nr. 116A  
200746, Craiova  
Dolj, ROMANIA

Description HCI544F1

Serial Number N19G304489

450-16383-G